

Lamola Boitumelo, 229246346
Maswanganyi Nsovo, 224788213
Assignment 2 Draft
Due Date: 23 March 2026

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SQL

1.

CREATE TABLE EMP_1 (
empNum **INTEGER** NOT NULL **UNIQUE**,
empLName **VARCHAR**(50),
empFName **VARCHAR**(50),
empInit **NCHAR**(10),
empHireDate **DATE**,
jobCode **INTEGER** NOT NULL,

PRIMARY KEY(empNum),
FOREIGN KEY(jobCode) **REFERENCES** JOB
);

2.

a)

1st insertion:

INSERT INTO EMP_1
VALUES ('101','News','John','G','2022-11-08','502');

2nd insertion:

INSERT INTO EMP_1
VALUES ('102','Senior','David','H','2021-07-12','501');

Results		Messages				
	empNum	empLName	empFName	empInit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501

Query executed successfully. localhost (17.0 RTM) BOITUMELO_KHANI\lamol ... Assign3DB 00:00:00 Row: 2, Col: 1 2 rows

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b)

```
INSERT INTO EMP_1
SELECT
empNum,
empLName,
empFName,
emplnit,
empHireDate,
jobCode
FROM Employee
WHERE empNum > 102;
```

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Results

Messages

	empNum	empLName	empFName	emplnit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501
3	103	Arbough	June	J	2017-12-01	503
4	104	Ramoras	Anne	AC	2009-11-15	501
5	105	Johnson	Alice	A	2014-02-01	502
6	106	Smithfield	William	W	2025-06-22	500
7	107	Alonzo	Maria	MM	2015-10-11	500
8	108	Washington	Ralph	RM	2013-08-22	501
9	109	Smith	Larry	LJ	2018-07-18	501
10	110	Olenko	Gerald	GB	2017-12-11	505
11	111	Wabash	Geoff	GG	2012-04-04	506
12	112	Smithson	Darlene	D	2016-10-23	507
13	113	Joebrood	Delbert	DD	2018-11-15	508
14	114	Jones	Annelise	AS	2016-08-20	508
15	115	Bawang	Cweka	CX	2013-01-25	501
16	116	Pratt	Gerald	G	2018-03-05	510
17	117	Williamson	Angie	A	2018-06-19	509
18	118	Frommer	James	J	2026-01-04	510
19	119	Smithfield	Will	WBG	2021-06-01	501

Query execu... | localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 19 rows

3.

```
UPDATE EMP_1
SET jobCode = 501
WHERE empNum = 107;
```

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	empNum	empLName	empFName	emplnit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501
3	103	Arbough	June	J	2017-12-01	503
4	104	Ramoras	Anne	AC	2009-11-15	501
5	105	Johnson	Alice	A	2014-02-01	502
6	106	Smithfield	William	W	2025-06-22	500
7	107	Alonzo	Maria	MM	2015-10-11	501
8	108	Washington	Ralph	RM	2013-08-22	501
9	109	Smith	Larry	LJ	2018-07-18	501
10	110	Olenko	Gerald	GB	2017-12-11	505
11	111	Wabash	Geoff	GG	2012-04-04	506
12	112	Smithson	Darlene	D	2016-10-23	507
13	113	Joebrood	Delbert	DD	2018-11-15	508
14	114	Jones	Annelise	AS	2016-08-20	508
15	115	Bawang	Cweka	CX	2013-01-25	501
16	116	Pratt	Gerald	G	2018-03-05	510
17	117	Williamson	Angie	A	2018-06-19	509
18	118	Frommer	James	J	2026-01-04	510
19	119	Smithfield	Will	WBG	2021-06-01	501

Query exec... | localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 7, Col: 1 | 19 rows

4.
DELETE FROM EMP_1
WHERE
empFName = 'William' AND
empLName = 'Smithfield' AND
empHireDate = '2025-06-22' AND
jobCode = 500;

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Results		Messages				
	empNum	empLName	empFName	empLName	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501
3	103	Arbough	June	J	2017-12-01	503
4	104	Ramoras	Anne	AC	2009-11-15	501
5	105	Johnson	Alice	A	2014-02-01	502
6	107	Alonzo	Maria	MM	2015-10-11	501
7	108	Washington	Ralph	RM	2013-08-22	501
8	109	Smith	Larry	LJ	2018-07-18	501
9	110	Olenko	Gerald	GB	2017-12-11	505
10	111	Wabash	Geoff	GG	2012-04-04	506
11	112	Smithson	Darlene	D	2016-10-23	507
12	113	Joebrood	Delbert	DD	2018-11-15	508
13	114	Jones	Annelise	AS	2016-08-20	508
14	115	Bawangi	Cweka	CX	2013-01-25	501
15	116	Pratt	Gerald	G	2018-03-05	510
16	117	Williamson	Angie	A	2018-06-19	509
17	118	Frommer	James	J	2026-01-04	510
18	119	Smithfield	Will	WBG	2021-06-01	501

Query executed successfully. Localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 5, Col: 1 | 18 rows

5.
SELECT *
INTO EMP_2
FROM EMP_1;

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The screenshot shows a SQL Server Enterprise Manager window with two tabs. The active tab is 'SQLQuery4.s... \lamol (80))' containing a SQL query. The query is a SELECT statement with a TOP clause, listing employee details from a table named EMP_2. The results pane below shows 18 rows of data. The status bar at the bottom indicates the query was executed successfully on the 'Assign3DB' database.

```
1 SELECT TOP (1000) [empNum]
2     , [emplName]
3     , [empFName]
4     , [empInit]
5     , [empHireDate]
6     , [jobCode]
7 FROM [Assign3DB].[dbo].[EMP_2]
8
```

	empNum	emplName	empFName	emplnit	empHireDate	jobCode
1	101	News	John	G	2022-11-08	502
2	102	Senior	David	H	2021-07-12	501
3	103	Arbough	June	J	2017-12-01	503
4	104	Ramoras	Anne	AC	2009-11-15	501
5	105	Johnson	Alice	A	2014-02-01	502
6	107	Alonzo	Maria	MM	2015-10-11	501
7	108	Washington	Ralph	RM	2013-08-22	501
8	109	Smith	Larry	LJ	2018-07-18	501
9	110	Olenko	Gerald	GB	2017-12-11	505
10	111	Wabash	Geoff	GG	2012-04-04	506
11	112	Smithson	Darlene	D	2016-10-23	507
12	113	Joenbrood	Delbert	DD	2018-11-15	508
13	114	Jones	Annelise	AS	2016-08-20	508
14	115	Bawangi	Cweka	CX	2013-01-25	501
15	116	Pratt	Gerald	G	2018-03-05	510
16	117	Williamson	Angie	A	2018-06-19	509
17	118	Frommer	James	J	2026-01-04	510
18	119	Smithfield	Will	WBG	2021-06-01	501

Query execute... | localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 18 rows

6.

ALTER TABLE EMP_2
ADD empPerc DECIMAL;

ALTER TABLE EMP_2
ADD projNum INT;

Can do single ALTER.... ADD

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	empNum	empLName	empFName	empLnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	NULL	NULL
2	102	Senior	David	H	2021-07-12	501	NULL	NULL
3	103	Arbough	June	J	2017-12-01	503	NULL	NULL
4	104	Ramoras	Anne	AC	2009-11-15	501	NULL	NULL
5	105	Johnson	Alice	A	2014-02-01	502	NULL	NULL
6	107	Alonzo	Maria	MM	2015-10-11	501	NULL	NULL
7	108	Washington	Ralph	RM	2013-08-22	501	NULL	NULL
8	109	Smith	Larry	LJ	2018-07-18	501	NULL	NULL
9	110	Olenko	Gerald	GB	2017-12-11	505	NULL	NULL
10	111	Wabash	Geoff	GG	2012-04-04	506	NULL	NULL
11	112	Smithson	Darlene	D	2016-10-23	507	NULL	NULL
12	113	Joebrood	Delbert	DD	2018-11-15	508	NULL	NULL
13	114	Jones	Annelise	AS	2016-08-20	508	NULL	NULL
14	115	Bawangi	Cweka	CX	2013-01-25	501	NULL	NULL
15	116	Pratt	Gerald	G	2018-03-05	510	NULL	NULL
16	117	Williamson	Angie	A	2018-06-19	509	NULL	NULL
17	118	Frommer	James	J	2026-01-04	510	NULL	NULL
18	119	Smithfield	Will	WBG	2021-06-01	501	NULL	NULL

✓ Quer... | localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 18 rows

7.

a) **UPDATE** EMP_2
SET empPerc = 4
WHERE empNum = 103;

	empNum	empLName	empFName	empLnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	NULL	NULL
2	102	Senior	David	H	2021-07-12	501	NULL	NULL
3	103	Arbough	June	J	2017-12-01	503	4	NULL
4	104	Ramoras	Anne	AC	2009-11-15	501	NULL	NULL
5	105	Johnson	Alice	A	2014-02-01	502	NULL	NULL
6	107	Alonzo	Maria	MM	2015-10-11	501	NULL	NULL
7	108	Washington	Ralph	RM	2013-08-22	501	NULL	NULL
8	109	Smith	Larry	LJ	2018-07-18	501	NULL	NULL

✓ Quer... | localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 3, Col: 7 | 18 rows

b) **UPDATE** EMP_2
SET empPerc = 5
WHERE empNum = 101
 OR empNum = 105
 OR empNum = 107;

There is a more optimal way

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Results		Messages							
	empNum	empLName	empFName	empInIt	empHireDate	jobCode	empPerc	projNum	
1	101	News	John	G	2022-11-08	502	5	NULL	
2	102	Senior	David	H	2021-07-12	501	NULL	NULL	
3	103	Arbough	June	J	2017-12-01	503	4	NULL	
4	104	Ramoras	Anne	AC	2009-11-15	501	NULL	NULL	
5	105	Johnson	Alice	A	2014-02-01	502	5	NULL	
6	107	Alonzo	Maria	MM	2015-10-11	501	5	NULL	
7	108	Washington	Ralph	RM	2013-08-22	501	NULL	NULL	
8	109	Smith	Larry	LJ	2018-07-18	501	NULL	NULL	

✓ Quer... | 🔒 localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 18 rows

- c) **UPDATE** EMP_2
SET empPerc = 10
WHERE empPerc IS NULL;

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Results		Messages							
	empNum	empLName	empFName	empInIt	empHireDate	jobCode	empPerc	projNum	
1	101	News	John	G	2022-11-08	502	5	NULL	
2	102	Senior	David	H	2021-07-12	501	10	NULL	
3	103	Arbough	June	J	2017-12-01	503	4	NULL	
4	104	Ramoras	Anne	AC	2009-11-15	501	10	NULL	
5	105	Johnson	Alice	A	2014-02-01	502	5	NULL	
6	107	Alonzo	Maria	MM	2015-10-11	501	5	NULL	
7	108	Washington	Ralph	RM	2013-08-22	501	10	NULL	
8	109	Smith	Larry	LJ	2018-07-18	501	10	NULL	
9	110	Olenko	Gerald	GB	2017-12-11	505	10	NULL	
10	111	Wabash	Geoff	GG	2012-04-04	506	10	NULL	
11	112	Smithson	Darlene	D	2016-10-23	507	10	NULL	
12	113	Joebrood	Delbert	DD	2018-11-15	508	10	NULL	
13	114	Jones	Annelise	AS	2016-08-20	508	10	NULL	
14	115	Bawangi	Cweka	CX	2013-01-25	501	10	NULL	
15	116	Pratt	Gerald	G	2018-03-05	510	10	NULL	
16	117	Williamson	Angie	A	2018-06-19	509	10	NULL	
17	118	Frommer	James	J	2026-01-04	510	10	NULL	
18	119	Smithfield	Will	WBG	2021-06-01	501	10	NULL	

✓ Quer... | 🔒 localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 18 rows

- d) **UPDATE** EMP_2
SET projNum = 25
WHERE jobCode >= 502;

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Results		Messages							
	empNum	empLName	empFName	emplnit	empHireDate	jobCode	empPerc	projNum	
1	101	News	John	G	2022-11-08	502	5	25	
2	102	Senior	David	H	2021-07-12	501	10	NULL	
3	103	Arbough	June	J	2017-12-01	503	4	25	
4	104	Ramoras	Anne	AC	2009-11-15	501	10	NULL	
5	105	Johnson	Alice	A	2014-02-01	502	5	25	
6	107	Alonzo	Maria	MM	2015-10-11	501	5	NULL	
7	108	Washington	Ralph	RM	2013-08-22	501	10	NULL	
8	109	Smith	Larry	LJ	2018-07-18	501	10	NULL	
9	110	Olenko	Gerald	GB	2017-12-11	505	10	25	
10	111	Wabash	Geoff	GG	2012-04-04	506	10	25	
11	112	Smithson	Darlene	D	2016-10-23	507	10	25	
12	113	Joebrood	Delbert	DD	2018-11-15	508	10	25	
13	114	Jones	Annelise	AS	2016-08-20	508	10	25	
14	115	Bawangi	Cweka	CX	2013-01-25	501	10	NULL	
15	116	Pratt	Gerald	G	2018-03-05	510	10	25	
16	117	Williamson	Angie	A	2018-06-19	509	10	25	
17	118	Frommer	James	J	2026-01-04	510	10	25	
18	119	Smithfield	Will	WBG	2021-06-01	501	10	NULL	

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e) **UPDATE** EMP_2
SET projNum = 14
WHERE empHireDate < '2021-07-10' **3**
AND jobCode <= 502;

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Results

Messages

	empNum	empLName	empFName	emplnit	empHireDate	jobCode	empPerc	projNum
1	101	News	John	G	2022-11-08	502	5	25
2	102	Senior	David	H	2021-07-12	501	10	NULL
3	103	Arbough	June	J	2017-12-01	503	4	25
4	104	Ramoras	Anne	AC	2009-11-15	501	10	14
5	105	Johnson	Alice	A	2014-02-01	502	5	14
6	107	Alonzo	Maria	MM	2015-10-11	501	5	14
7	108	Washington	Ralph	RM	2013-08-22	501	10	14
8	109	Smith	Larry	LJ	2018-07-18	501	10	14
9	110	Olenko	Gerald	GB	2017-12-11	505	10	25
10	111	Wabash	Geoff	GG	2012-04-04	506	10	25
11	112	Smithson	Darlene	D	2016-10-23	507	10	25
12	113	Joebrood	Delbert	DD	2018-11-15	508	10	25
13	114	Jones	Annelise	AS	2016-08-20	508	10	25
14	115	Bawangi	Cweka	CX	2013-01-25	501	10	14
15	116	Pratt	Gerald	G	2018-03-05	510	10	25
16	117	Williamson	Angie	A	2018-06-19	509	10	25
17	118	Frommer	James	J	2026-01-04	510	10	25
18	119	Smithfield	Will	WBG	2021-06-01	501	10	14

✓ Quer... | localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 18 rows

f) `SELECT empNum, empLName, empFName`
`FROM EMP_2`
`WHERE empLName LIKE 'Smith%'`
`ORDER BY empLName DESC;`

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Results		Messages	
	empNum	empLName	empFName
1	112	Smithson	Darlene
2	119	Smithfield	Will
3	109	Smith	Larry

✓ Qu... | localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 3 rows

8.
`SELECT Employee.empNum,`
`empLName,`
`SUM (assignHrs) AS sumHRS,`
`SUM (assignChrg) AS sumCHRG`
`FROM Employee JOIN Assignment ON Employee.empNum = Assignment.empNum`
`GROUP BY Employee.empNum, empLName;`

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Results		Messages		
	empNum	empLName	sumHRS	sumCHRG
1	101	News	3	387.50
2	103	Arbough	20	1664.65
3	104	Ramoras	12	1218.70
4	105	Johnson	9	931.00
5	108	Washington	8	840.15
6	113	Joebrood	4	192.85
7	115	Bawangi	13	1276.75
8	117	Williamson	18	649.54

BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 8 rows

9.
`SELECT projNum,
SUM (assignHrs) AS sumHRS,
SUM (assignChrg) AS sumCHRG
FROM Assignment
GROUP BY projNum
ORDER BY projNum DESC;`

Results		Messages	
	projNum	sumHRS	sumCHRG
1	25	19	1668.16
2	22	28	2593.16
3	18	24	1544.80
4	15	20	1806.52

localhost (17.0 RTM) | BOITUMELO_KHANI\lamol ... | Assign3DB | 00:00:00 | Row: 1, Col: 1 | 4 rows